

MARINE DIGITAL TWIN PLATFORM

Offshore Wind Simulation & Environmental Impact System

DEVELOPMENT PLAN

Region	Scotian Shelf — Nova Scotia, Canada
Coordinates	43.68°N, 64.33°W → 44.83°N, 61.94°W
Primary Users	Researchers · Government · Investors
Windmill Types	Grounded & Floating Offshore Turbines
Data Sources	16 Scientific APIs · 169 Environmental Variables
Future Extension	Underwater Data Centers

Confidential — Prepared for internal use by research, government, and investment stakeholders

1 Platform Overview

The Marine Digital Twin Platform is a research-grade simulation environment that allows researchers, government bodies, and investment stakeholders to evaluate offshore wind turbine sites using exclusively real, verified oceanographic and atmospheric data. Users place a proposed turbine at any coordinate within the Scotian Shelf region and run a suite of scientific simulation modules to understand environmental impact, ecosystem risk, governance compliance, and financial viability — before any physical deployment occurs.

All computations are driven by a fused data cube built from 16 live scientific APIs, containing 169 environmental variables spanning ocean physics, wave dynamics, atmospheric conditions, biogeochemistry, species biology, seafloor characteristics, human activity, and governance layers. No synthetic or fabricated data is used at any point.

Component	Description
Study Region	Scotian Shelf — 43.68°N to 44.83°N, 64.33°W to 61.94°W
Spatial Grid	13 × 28 cells at 1/12° resolution (~8 km), 50 vertical depth levels
Temporal Coverage	1993-01-01 to present — reanalysis to 2023, NRT from 2024
Variable Catalogue	169 variables across 8 domain groups
Windmill Types	Grounded (monopile/jacket) and floating (spar/semi-sub)
Simulation Modules	8 independent modules — wake, noise, scour, Lagrangian, acoustic, species, conflict, o
Target Users	Researchers · Government regulators · Investors/Stakeholders
Future Extension	Underwater data center siting — same platform architecture

2 System Architecture

Data Cube Layer

A pre-built, static Zarr data store fusing all 16 real-data sources into a common 1/12° grid. Every cell carries a quality flag (direct observation · multi-model · single-model · interpolated · temporally aligned · missing). Missing data is left as NaN — never filled synthetically. The cube is updated on a scheduled basis via automated ingestion workflows.

Simulation Engine (Backend)

A Python-based FastAPI service that exposes RESTful endpoints for region metadata, site validation, tool discovery, and async job submission. Long-running simulations are dispatched to a Celery/Redis job queue. Results are stored as GeoJSON, JSON grids, and PNG figures on object storage and referenced by signed URLs.

Frontend Application

A React 18 + TypeScript single-page application centred on a 3D interactive globe (CesiumJS). Four floating configuration panels overlay the globe, allowing users to configure variables, simulation methods, turbine

specifications, and time ranges without leaving the map view. Simulation results are rendered as live overlays on the globe.

Automation Layer (n8n)

An n8n instance handles report generation, job completion notifications, scheduled near-real-time data refreshes, and optional government portal submissions. The Claude API is called within the report workflow to write executive summaries and data provenance narratives from structured simulation outputs.

3 Phase 1 — Data Infrastructure & Backend Foundation

All frontend development is blocked until the core data cube and API layer are operational. Phase 1 delivers the full Zarr cube, quality-flagged data pipeline, and a FastAPI skeleton with the six primary endpoint groups. Simulation Modules A, B1, and C1 are the minimum viable set needed to demonstrate end-to-end functionality.

Data Ingestion

BACKEND	Ingest Copernicus Global PHY reanalysis (GLORYS12) — 3D temperature, salinity, currents (u, v) at 1/12° resolution, 50 depth levels, 1993 to 2023
BACKEND	Ingest Copernicus NRT physics product — extend reanalysis to present, append hourly as new data arrives
BACKEND	Ingest HYCOM OPeNDAP — 3D T, S, U, V as secondary ensemble member for multi-model spread calculation
BACKEND	Ingest Copernicus Global Wave Analysis (WAV) — significant wave height, peak period, direction, Stokes drift components
BACKEND	Ingest ERA5 via CDS — wind (10 m and 100 m), surface pressure, heat flux, evaporation minus precipitation at 0.25°
BACKEND	Ingest Open-Meteo Marine + Atmosphere — waves and wind as third ensemble member, no authentication required
BACKEND	Ingest GEBCO 2026 bathymetry — 15 arc-second resolution, conservative-averaged to 1/12° common grid
BACKEND	Ingest DFO WebTide harmonic constituents — tidal current prediction for 10 constituents at Scotian Shelf
BACKEND	Ingest CIOOS Atlantic ERDDAP — buoy time series (waves, wind, SST, current profiles) and CTD cast profiles
BACKEND	Ingest Bedford Basin Monitoring Program (BBMP) — weekly CTD profiles for biogeochemical validation
BACKEND	Ingest Copernicus BGC model — chlorophyll, oxygen, nitrate, phosphate, silicate, pH, NPP at 0.25°
BACKEND	Ingest satellite ocean colour (CMEMS) — surface chlorophyll at ~1 km for higher-resolution surface BGC
BACKEND	Ingest OBIS — 501,900 species occurrence records in the ROI; grid to cell-level occurrence counts and species lists
BACKEND	Ingest Ocean Tracking Network (OTN) acoustic detections — real animal movement data for biological validation

BACKEND	Ingest Global Fishing Watch (GFW) 4Wings — gridded vessel presence at 0.01°–0.1° for human activity layer
BACKEND	Ingest DFO MPA boundaries, NRCan offshore lease blocks, CHS shipping lane shapefiles — vectorise to fractional grid coverage

Data Cube Construction

BACKEND	Define common grid specification: lat 43.68 to 44.83, lon -64.33 to -61.94, 1/12° step, 50 depth levels, hourly time axis
BACKEND	Regrid all sources to common grid using conservative interpolation — preserve mass and volume integrals
BACKEND	Compute multi-model ensemble mean and spread (std dev) for all variables available from 2+ sources
BACKEND	Assign quality flags 1–6 per cell per variable per timestep (direct · multi-model · single-model · interpolated · temporally aligned · missing)
BACKEND	Write final cube as Zarr store with chunking optimised for time-series extraction and spatial slicing
BACKEND	Generate metadata.json — source versions, build date, API access timestamps, ROI bounds, variable catalog
BACKEND	Implement cube update pipeline — append new NRT data from Copernicus and Open-Meteo on scheduled runs

API Layer — Core Endpoints

BACKEND	GET /api/region — return ROI bounds, temporal coverage dates, variable summary with domain group counts
BACKEND	GET /api/site/validate?lon=■= — depth lookup, data availability check, nearest buoy/CTD distances, MPA governance flags
BACKEND	GET /api/variables — full 169-variable catalogue with IDs, domain groups, source provenance, quality flag distributions
BACKEND	GET /api/tools — list all available simulation tools with category, tier, short description, and runtime estimates
BACKEND	GET /api/tools/{tool_id} — full tool metadata: required/optional variables, configurable parameters, expected output types
INFRA	POST /api/jobs — validate inputs, dispatch to Celery queue, return job_id and poll interval immediately
INFRA	GET /api/jobs/{job_id} — return job status, progress counters, estimated completion, or full result on completion

INFRA

GET /api/jobs/{job_id}/result/* — serve result files (GeoJSON trajectories, density grids, PNG figures) via signed object-storage URLs

Simulation Module A — Baseline Characterisation

BACKEND

Extract all selected variables at the site coordinate and surrounding ROI cells for the requested time window

BACKEND

Compute monthly climatology, seasonal means, inter-annual trends, and extreme-event statistics for each variable

BACKEND

Generate data quality report — flag any variables with >20% missing coverage, list nearest alternative data sources

BACKEND

Output: time series JSON, climatology tables, quality summary — consumed by all downstream modules

Simulation Module B1 — Wind Wake Deficit

BACKEND

Implement Jensen top-hat wake model using real 100 m wind speed and direction from ERA5 at the site

BACKEND

Implement Bastankhah Gaussian wake model as higher-fidelity alternative; expose model choice as user parameter

BACKEND

Compute wake decay coefficient alpha from atmospheric boundary layer height and surface roughness (real ERA5 variables)

BACKEND

Output: 2D spatial field of wind speed deficit (%) at hub height; JSON grid + PNG figure

Simulation Module C1 — Lagrangian Particle Tracking

BACKEND

Implement RK4 time integration for particle advection using 3D velocity fields (u_0 , v_0) from the cube

BACKEND

Add Stokes drift (VSDX, VSDY) from Copernicus WAV for surface-layer particles

BACKEND

Predict tidal currents at the site using DFO WebTide harmonic constituents (10 constituents, pure physics)

BACKEND

Compute vertical diffusivity K_z from real T and S stratification (N^2 profile) via K-profile parameterisation (Large et al. 1994)

BACKEND

Implement random-walk vertical diffusion step; check bathymetry for particle beaching

BACKEND

Compute output statistics: mean displacement, max displacement, residence time, dispersion ellipse, connectivity matrix

BACKEND

Output: GeoJSON LineString trajectories (one per particle), density grid, connectivity matrix, scalar stats, PNG figures

4 Phase 2 — Globe UI & Floating Configuration Panels

The primary workspace page — a full-viewport 3D interactive globe with four non-blocking floating panels. Development can begin against mock API responses in parallel with Phase 1 backend work. The globe renders the Scotian Shelf region with custom ocean colour theming and supports bounding-box environment selection, windmill pin placement, and real-time site validation feedback.

3D Globe Implementation

FRONTEND	Integrate CesiumJS globe with dark navy ocean colour theme matching platform aesthetic
FRONTEND	Configure custom bathymetry terrain layer using GEBCO data with blue-gradient depth colouring
FRONTEND	Load Scotian Shelf ROI boundary from /api/region on page load; constrain all interactions to this region
FRONTEND	Implement GLSL ocean surface shader — depth-based teal-to-navy gradient that matches real bathymetry data
FRONTEND	Add globe rotation, zoom, and tilt controls with momentum; lock to North Atlantic view on load
FRONTEND	Display coastline, shipping lane overlays, and MPA boundary polygons as vector layers from governance data

Region Selection Tools

FRONTEND	Implement bounding box draw tool — click-drag to define rectangular environment region on globe surface
FRONTEND	Show coordinate readout during draw; snap to grid cells for alignment with the 1/12° cube resolution
FRONTEND	Provide four manual coordinate input fields (SW lat/lon, NE lat/lon) with real-time globe update on change
FRONTEND	Highlight selected region with semi-transparent fill and dashed border; show grid cell count overlay
FRONTEND	Clear/redraw button to reset region without losing windmill placement

Windmill Placement & Validation

FRONTEND	Implement windmill pin drag-to-place on globe surface within selected region
FRONTEND	Fire GET /api/site/validate on every pin move (debounced 300 ms); show loading indicator on pin
FRONTEND	Display site validation result as a compact overlay card near the pin: water depth, data quality badges, nearest buoy distance
FRONTEND	Show governance alerts if pin is inside or within 10 km of an MPA — yellow warning banner

FRONTEND	Provide one manual coordinate input for windmill position (lat/lon) with immediate globe pan and validation
FRONTEND	Windmill type toggle — Grounded vs Floating — updates turbine specification defaults in Panel C

Floating Panel A — Variable Selector

FRONTEND	Build collapsible floating panel anchored to left edge of viewport
FRONTEND	Load variable catalogue from GET /api/variables; group into 8 domain accordions (Physics, Waves, Atmosphere, BGC, Biology, Seafloor, Human Activity, Governance)
FRONTEND	Implement domain-level dropdown with Select All / Deselect All per group
FRONTEND	Implement individual variable checkboxes with source quality badge (multi-model · single-model · in-situ)
FRONTEND	Show data availability warning (yellow) for variables with missing coverage at the selected site
FRONTEND	Show selected variable count badge on panel header; cap enforced at tool's required + optional variable list

Floating Panel B — Analysis Methods

FRONTEND	Build collapsible floating panel anchored to right edge of viewport
FRONTEND	Load tool list from GET /api/tools; display as grouped cards by category (physics simulation, statistical, ecological, economic)
FRONTEND	On tool selection, call GET /api/tools/{tool_id} and render dynamic parameter form for that tool's configurable inputs
FRONTEND	Show estimated runtime label per tool based on current particle count / time range settings
FRONTEND	Support multi-tool selection — show stacked parameter forms; validate inter-tool variable dependencies
FRONTEND	Tool cards: MCMC Bayesian Inference, Lagrangian Particle Tracking, Statistical Inference, Correlation Analysis, Wind Wake Model, Acoustic Propagation, Species Risk Assessment, Multi-objective Optimisation

Floating Panel C — Turbine Specifications

FRONTEND	Build compact floating panel (bottom-left) with turbine engineering parameter inputs
FRONTEND	Fields: hub height (m), rotor diameter (m), rated power (MW), column diameter (m), foundation type, floater radius (m — floating only), draft depth (m — floating only)
FRONTEND	Pre-populate with defaults on windmill type toggle (Grounded: 150 m hub, 236 m rotor, 15 MW; Floating: adjusted defaults)
FRONTEND	All fields passed directly to simulation job payload on submit

Floating Panel D — Time Range

FRONTEND	Build slim floating panel (bottom of viewport) with a year-range slider
FRONTEND	Slider bounds populated from temporal_coverage in /api/region response (1993 to present)
FRONTEND	Separate markers for reanalysis end (2023) and NRT start (2024) — visual indicator on slider track
FRONTEND	Selected range displayed as 'YYYY – YYYY' label with approximate data volume estimate

Submit Flow & Job Polling

FRONTEND	Submit button: validate all panels complete → POST /api/jobs → show job queued toast with job ID
FRONTEND	Progress overlay on globe: animated spinner with live status text from polling response (timesteps, particles beached)
FRONTEND	Progress bar component: percent complete, active particle count, estimated seconds remaining
FRONTEND	On completion: transition globe to results overlay mode; enable Generate Report button
FRONTEND	Error state handling: surface API error messages in a dismissible alert banner above the globe

5 Phase 3 — Simulation Visualisation on Globe & Remaining Backend Modules

Once the job polling returns a completed result, simulation outputs are rendered as interactive overlays directly on the 3D globe within the selected region. Each overlay type uses a dedicated deck.gl layer for GPU-accelerated rendering. Concurrently, the remaining six simulation modules are implemented in the backend.

Globe Overlay Rendering

FRONTEND	Lagrangian particle trajectories — render GeoJSON LineStrings as animated TripsLayer (deck.gl) with time scrubber
FRONTEND	Particle density heatmap — render density.json as HeatmapLayer with semi-transparent warm colour ramp
FRONTEND	Connectivity matrix — render grid as GridCellLayer coloured by particle arrival probability (0–100%)
FRONTEND	Wind wake deficit — render wake field as BitmapLayer or GridCellLayer showing wind speed reduction (%)
FRONTEND	Acoustic noise contours — render noise propagation field as ContourLayer at key dB thresholds
FRONTEND	Species risk map — render per-group risk gradient as HeatmapLayer with toggleable species groups
FRONTEND	Cumulative impact score — single composite layer showing integrated multi-stressor footprint
FRONTEND	Shipping/fishing conflict — render GFW vessel density and conflict probability as separate toggleable layers

Animation & Layer Controls

FRONTEND	Play / Pause / Step controls for trajectory animation with speed multiplier (1×, 5×, 20×)
FRONTEND	Time scrubber slider (hour 0 to simulation end) with current timestamp label
FRONTEND	Layer switcher panel — toggle each overlay type on/off; adjust opacity per layer
FRONTEND	Legend component — auto-generated colour scale for active overlay with units and min/max values
FRONTEND	Windmill site marker — 3D pin or turbine icon on globe at simulation coordinate

Backend Module B2 — Underwater Noise Source

BACKEND	Compute operational turbine noise source level using Tougaard et al. (2020) measured coefficients from real wind farms
BACKEND	Compute construction pile-driving noise using Bailey et al. (2010) — time-limited impact phase

BACKEND

Compute ambient noise baseline from Wenz (1962) curves using real significant wave height and wind speed

BACKEND

Output: noise source level (dB re 1 μ Pa at 1 m) at key frequencies (10 Hz, 100 Hz, 1 kHz)

Backend Module B3 — Foundation Scour**BACKEND**

Compute combined wave-current bed shear stress using Soulsby (1997) formulation from real wave height, period, and bottom current

BACKEND

Estimate maximum scour depth using Sumer and Fredsøe (2002) from Shields parameter and grain size (where sediment data available)

BACKEND

Flag locations where sediment data is unavailable (quality flag 6) — report as data gap, not estimated value

BACKEND

Output: bed shear stress time series, scour depth scalar, uncertainty range

Backend Module C2 — Acoustic Propagation**BACKEND**

Compute sound speed profile from real T, S, depth using Chen-Millero 1977 equation (UNESCO 1983)

BACKEND

Compute frequency-dependent absorption coefficient using François-Garrison (1982) from real T, S, pH, depth

BACKEND

Implement transmission loss calculation propagating noise source outward from turbine foundation

BACKEND

Output: 2D noise level map (dB) at key frequencies; range-to-threshold distances (120 dB, 160 dB, 180 dB)

Backend Module C3 — Species Risk Assessment**BACKEND**

Load OBIS species occurrence grid for the ROI — real occurrence records, not modelled distributions

BACKEND

Map acoustic noise footprint against species hearing sensitivity bands (fish, marine mammals, invertebrates)

BACKEND

Map Lagrangian connectivity footprint against known spawning and nursery areas from OBIS seasonal data

BACKEND

Compute habitat suitability change score from delta-chlorophyll, delta-temperature, and wake-induced upwelling

BACKEND

Output: per-species-group risk spatial field, species list with risk rating, habitat change map

Backend Module C4 — Cumulative Impact Score**BACKEND**

Normalise all Module B and C output layers to 0–1 scale using real observed range at the site

BACKEND	Apply user-specified or default weighting vector across impact categories
BACKEND	Compute cell-wise weighted sum across all active impact layers for the ROI grid
BACKEND	Output: 2D cumulative impact score grid, stacked bar chart of contributions per cell, top-5 impact drivers

Backend Module D — Human Conflict Analysis

BACKEND	Compute vessel transit conflict score from GFW vessel presence density at site and downwind/downstream area
BACKEND	Compute fishing conflict from GFW fishing vessel hours in the ROI — flag protected or seasonal fishing grounds
BACKEND	Check site against DFO MPA buffers (500 m, 1 km, 5 km) and NRCan lease block boundaries
BACKEND	Check site against CHS shipping lane centre lines — compute lateral clearance distance
BACKEND	Output: conflict probability scores per sector, governance compliance report, minimum clearance distances

Backend Module E — Multi-objective Site Optimisation

BACKEND	Define objective functions: maximise wind energy (from ERA5 100 m wind), minimise cumulative impact (Module C4), minimise human conflict (Module D)
BACKEND	Run NSGA-II multi-objective genetic algorithm (Deb et al. 2002) across all valid grid cells in the ROI
BACKEND	Identify Pareto-optimal sites — coordinates where no objective can improve without another worsening
BACKEND	Rank all Pareto sites by user-specified priority weighting (energy vs ecology vs conflict)
BACKEND	Output: ranked site table (lat/lon, energy score, impact score, conflict score), Pareto front scatter data

Backend Module F — MCMC Uncertainty Quantification

BACKEND	Define prior distributions over uncertain inputs using real data spread (multi-model Copernicus vs HYCOM ensemble)
BACKEND	Set up likelihood function comparing model predictions against real buoy observations at the site
BACKEND	Run MCMC sampling (via PyMC or emcee) to estimate posterior distributions of key simulation outputs
BACKEND	Compute Gelman-Rubin R-hat convergence diagnostic; flag non-converged chains
BACKEND	Data-driven model weighting — use MCMC to estimate which data source best matches real buoy observations

BACKEND

Output: posterior PDF per key scalar, 95% credible intervals, trace plots, R-hat diagnostics

6 Phase 4 — Results Page & ROI Dashboard

Two dedicated output pages that present simulation results in audience-appropriate formats. The Environmental Impact Results page targets researchers and government bodies with scientific depth and full data provenance. The ROI Dashboard targets investors and stakeholders with financial projections, site rankings, and risk-adjusted return analysis.

Environmental Impact Results Page

FRONTEND	Page layout: sticky section navigation (Executive Summary · Physical Footprint · Ecosystem · Lagrangian · Human Conflict · Uncertainty · Data Provenance)
FRONTEND	Executive Summary section: six scalar metric cards (mean displacement, max displacement, residence time, particles beached, cumulative impact score, dominant impact driver)
FRONTEND	Physical Footprint section: side-by-side small maps for wake deficit field, acoustic noise contours, scour depth estimate, EMF profile
FRONTEND	Ecosystem Impact section: species risk table (group, risk level, affected area km ² , key species list), habitat suitability change map
FRONTEND	Lagrangian section: trajectory map PNG from backend, displacement time series chart (mean $\pm$ 1 σ , median, p5/p95), dispersion ellipse summary card
FRONTEND	Human Conflict section: MPA compliance summary (nearest MPA, distance, inside/outside), shipping lane clearance, fishing conflict score by season
FRONTEND	Uncertainty section: MCMC posterior density plots (Plotly), credible interval summary table, R-hat convergence badges, data source disagreement highlights
FRONTEND	Data Provenance accordion: per-variable source list, quality flags, model versions, DOI citations, date of last data refresh
FRONTEND	Compare Sites feature — overlay results from two saved simulation jobs side by side
FRONTEND	Download section: JSON results, PNG figures, GeoJSON trajectories, full provenance manifest as individual download links

ROI Dashboard Page

FRONTEND	Page layout: two-column dashboard — left panel inputs, right panel live-updating output charts
FRONTEND	Energy Generation section: annual MWh projection from wake-adjusted wind resource, capacity factor by month (bar chart), wind rose showing directional energy breakdown
FRONTEND	Financial Inputs form: capex (£/MW), opex (£/MWh/year), energy sale price (£/MWh), discount rate (%), project lifetime (years) — all recalculate on change
FRONTEND	ROI output cards: NPV, IRR, simple payback period, levelised cost of energy (LCOE) — all with uncertainty ranges from MCMC

FRONTEND	Annual cashflow waterfall chart (D3) — capex year 0, revenue stream, opex deductions, cumulative NPV line
FRONTEND	Site Ranking table: Pareto-optimal sites from Module E, sortable by energy / impact / conflict / combined score
FRONTEND	Pareto front scatter plot (D3): energy score vs cumulative impact score, user's site highlighted, other Pareto sites as interactive points
FRONTEND	Risk-adjusted return section: tornado chart showing sensitivity of NPV to each uncertain input (wind variability, energy price, capex overrun)
FRONTEND	Export: investor-facing PDF report (triggers n8n workflow), Excel financial model download
Shared Result Infrastructure	
FRONTEND	Shareable simulation URL — encode job ID and configuration hash in URL; recipients view read-only results without reconfiguring
FRONTEND	Saved Simulations page — list of past jobs with site coordinates, date, key metrics, link to results and report
BACKEND	GET /api/roi/{job_id}?capex=&price;=&discount;= — server-side ROI calculation returning NPV, IRR, LCOE, cashflows
BACKEND	POST /api/saved-simulations — persist job ID, input configuration, and summary metrics to user account
BACKEND	GET /api/jobs/{job_id}/report — trigger n8n report workflow, return status and download URL when ready

7 Phase 5 — Integrations, Authentication & Production Readiness

Final phase covering automation workflows, user authentication with role-based access, performance optimisation, accessibility compliance, and the groundwork for the underwater data center extension. All error states, loading states, and empty states are hardened across the full application.

n8n Automation Workflows

INTEGRATION	Report Generation workflow: on trigger, pull all module outputs from job → assemble report sections → call Claude API for executive summary narrative → compile figures → generate PDF → store on object storage → return download URL
INTEGRATION	Investor Report variant: same workflow with financial section prepended, technical jargon suppressed, branded header/footer
INTEGRATION	Job Completion Notification workflow: poll job status every 60 s → on completion, send email or Slack message with results link and key metrics summary
INTEGRATION	Near-Real-Time Data Refresh workflow: hourly trigger → fetch latest Copernicus NRT and Open-Meteo data → append to Zarr cube → update metadata.json latest date
INTEGRATION	Government Portal Submission workflow (optional): format simulation result as structured environmental pre-screening payload → POST to DFO online portal on user request

Authentication & Authorisation

INFRA	Implement JWT-based authentication with refresh tokens
INFRA	Three user roles: Researcher (full access to all panels, tools, raw data exports), Government (full access + governance compliance highlights), Investor/Viewer (ROI dashboard + report download only — no raw simulation controls)
INFRA	Role-based UI gating — Investor role sees simplified globe view with windmill placement and ROI page; panels B and C hidden
INFRA	Organisation accounts — multiple users per organisation, shared saved simulations within org
INFRA	Audit log — record all simulation submissions with user, timestamp, coordinates, selected tools for compliance purposes

Performance & Scalability

INFRA	Lazy-load CesiumJS globe tiles — only fetch terrain and imagery tiles for current viewport extent
FRONTEND	Stream GeoJSON trajectory data in chunks — render particles progressively as trajectory file downloads
INFRA	Cache /api/region, /api/variables, /api/tools responses at CDN edge — these change only on cube rebuild

INFRA	Celery worker auto-scaling — add workers when job queue depth exceeds threshold; scale down on idle
INFRA	Pre-compute frequently-requested baselines (top 10 grid cells by validation request volume) as warm cache
BACKEND	Implement request coalescing — if two users submit identical jobs within 5 minutes, return same result

Accessibility & Cross-browser

FRONTEND	WCAG 2.1 AA audit across all three pages — colour contrast, focus indicators, screen reader labels
FRONTEND	Full keyboard navigation for all four floating panels — open/close, accordion expand, checkbox toggle, slider control
FRONTEND	ARIA live regions for simulation progress updates — screen readers announce completion without visual polling
FRONTEND	Test and fix layout across Chrome, Firefox, Safari, Edge; test on iPad (tablet layout for floating panels)
FRONTEND	Responsive floating panel layout — collapse to bottom drawer on screens narrower than 1024 px

Error States & Data Quality UX

FRONTEND	Globe: show hatched overlay on cells where selected variables have missing data (quality flag 6)
FRONTEND	Site validation: show yellow alert when sediment data unavailable; offer to proceed with degraded scour estimate
FRONTEND	Job failure: parse error type from API response — show actionable message (e.g. 'Reduce particle count or shorten time window to fit within runtime limit')
FRONTEND	Data provenance: highlight in red any variable where model spread exceeds 20% of mean — flag as high uncertainty
FRONTEND	Empty state for Saved Simulations page — clear call-to-action directing user to globe to start first simulation

Underwater Data Center Extension Groundwork

INFRA	Add 'site type' selector to globe page (Offshore Wind Turbine · Underwater Data Center) — routes to appropriate module set
BACKEND	Scope new simulation modules for data center siting: thermal plume impact, cooling water intake/discharge, cable routing conflict, acoustic signature
BACKEND	Identify additional variables needed for data center siting — deep-water temperature stability, current speed at depth, bathymetric slope

INFRA

Confirm existing cube variables cover data center requirements; schedule additional ingestion if gaps identified

8 API Contract Reference

All frontend rendering is driven exclusively by these API responses. The frontend team never accesses the data cube directly. The contract below defines the current agreed endpoints plus the additional endpoints required to support the full UI.

Method	Endpoint	Purpose	Consumed By
GET	/api/region	ROI bounds, temporal coverage, variable summary	Globe load, date picker, Panel A
GET	/api/site/validate	Depth, data availability, nearest buoy/CTD, MPA governance	Site placement, site feedback card
GET	/api/variables	Full 169-variable catalogue with groups, sources, quality flags	Panel A checkbox tree
GET	/api/tools	All simulation tools — category, tier, runtime estimates	Panel B tool cards
GET	/api/tools/{tool_id}	Required/optional variables, configurable parameters, output types	Panel B dynamic parameter form
POST	/api/jobs	Submit simulation job — returns job_id and poll interval	Submit button
GET	/api/jobs/{job_id}	Job status, progress %, active particles, full result on completion	Polling, progress bar, overlay render
GET	/api/jobs/{job_id}/result/*	GeoJSON trajectories, density grids, PNG figures via signed URLs	Global overlays, Results page
GET	/api/roi/{job_id}	NPV, IRR, LCOE, cashflows given user financial inputs	ROI Dashboard
POST	/api/saved-simulations	Persist job config and summary to user account	Save simulation button
GET	/api/jobs/{job_id}/report	Trigger n8n report, return download URL on completion	Generate Report button

9 Recommended Technology Stack

Layer	Technology	Purpose
Globe & Mapping	CesiumJS	3D globe, terrain, custom GLSL ocean shaders
Globe & Mapping	deck.gl	GPU-accelerated overlays — heatmap, trajectories, grid cells
Globe & Mapping	Turf.js	Client-side coordinate math, bounding box operations
Frontend Framework	React 18 + TypeScript	Component architecture, type safety
State Management	Zustand	Global simulation state (panels, job, results, overlays)
Data Fetching	React Query (TanStack)	API polling, caching, background refetch for job status
Charts	Recharts	Time series, displacement plots, capacity factor bar charts
Charts	D3.js	Pareto scatter, tornado chart, custom dispersion ellipse
Charts	Plotly	MCMC trace plots, posterior density, interactive zoom
Backend	FastAPI (Python)	Async REST API, OpenAPI schema generation
Job Queue	Celery + Redis	Async simulation dispatch, worker auto-scaling
Data Storage	Zarr (xarray)	Fused data cube — chunked, compressed, cloud-ready
Object Storage	S3-compatible (MinIO / AWS S3)	GeoJSON results, PNG figures, generated PDF reports
Database	PostgreSQL	User accounts, saved simulations, audit log

Automation	n8n	Report generation, notifications, data refresh workflows
Auth	JWT + refresh tokens	Role-based access — Researcher, Government, Investor
Infrastructure	Docker Compose (dev) / K8s (prod)	Containerised services, Celery worker scaling

10 Task Tag Reference

Tag	Colour	Meaning
BACKEND	Teal	Python simulation code, Zarr cube, FastAPI endpoints, scientific modules
FRONTEND	Blue	React components, CesiumJS globe, deck.gl overlays, UI panels, charts
INFRA	Purple	Docker, Celery/Redis, PostgreSQL, S3, authentication, deployment, scaling
INTEGRATION	Amber	n8n automation workflows, Claude API calls, external portal submissions